

Energy Data Science Intern Problem

Overview

Dairyland Power Cooperative is looking forward to your upcoming interview. As part of the interview process, we ask you to complete a short technical exercise that highlights your data acquisition, analysis, and modeling skills.

Please complete the exercise before your interview and be prepared to present your findings during the interview. You may use any format that works well for your results, such as PowerPoint, Jupyter Notebook, HTML, or a PDF report. We expect the exercise to take no more than four hours.

Instructions

1. Timing and Submission

- You have received this problem in advance, so you have time to prepare.
- Please email your presentation materials to David Elzinga (david.elzinga@dairylandpower.com) at least 2 hours before your interview.
- Plan for a 10-minute presentation, followed by discussion with the interview team.

2. Ownership and Resources

- This is individual work. Please do not share the problem statement or ask another person to complete any part of the exercise for you.
- You may use class notes, textbooks, online documentation, and AI tools to support your work.
- You should fully understand your methodology and results and be ready to explain them independently during the interview.

3. Deliverable Expectations

Your presentation should:

- Clearly explain your approach, analysis, and key findings.
- Be understandable to both technical and non-technical interviewers.
- Use visuals, charts, or tables when they help communicate your results.

4. Technical Notes

- You may use any programming language.
- During the interview, we may ask about your reasoning, assumptions, limitations, and possible improvements.

Background

Dairyland Power Cooperative (DPC) is a member of the Midcontinent Independent System Operator (MISO). MISO manages electricity flow and pricing across 15 U.S. states and Manitoba, Canada.

One common energy price model is the Locational Marginal Price (LMP). LMP is the price paid to a generator a generator (such as a Dairyland) to generate one additional megawatt (MW) of power at a specific location, called a node. LMP varies by location and time because demand, grid conditions, and generation costs change.

Task

Consider DPC's hydroelectric plant, Flambeau, in Ladysmith, Wisconsin. Flambeau is paid the LMP for each MW it produces. For example, if Flambeau produces 3 MW in one hour and the LMP is \$40/MW, the revenue for that hour is $3 \text{ MW} \times \$40/\text{MW} = \120 .

Suppose DPC is planning a continuous 2-week maintenance outage in 2027, during which Flambeau will produce zero MW. Your task is to recommend the best time for that outage. In this problem, "best" means a period when the opportunity cost, or lost revenue from being offline, is *consistently* low.

Assume there are no operational, weather, or other constraints that prevent DPC from choosing any 2-week period in 2027.

Data

Historical LMP prices are publicly available through MISO's Market Reports portal. Use the following guidance to find the data:

1. Open the MISO Market Reports page: [https://www.misoenergy.org/markets-and-operations/real-time--market-data/market-reports/#nt=%2FMarketReportType%3AHistorical%20LMP%2FMarketReportName%3ADay-Ahead%20Market%20ExPost%20LMPs%20\(csv\)&t=10&p=0&s=MarketReportPublished&sd=desc](https://www.misoenergy.org/markets-and-operations/real-time--market-data/market-reports/#nt=%2FMarketReportType%3AHistorical%20LMP%2FMarketReportName%3ADay-Ahead%20Market%20ExPost%20LMPs%20(csv)&t=10&p=0&s=MarketReportPublished&sd=desc)
2. Confirm the highlighted folder on the left is **Day-Ahead Market ExPost LMPs (csv)**. You do not need to know what "Day-Ahead" or "ExPost" means to complete the exercise.
3. Daily files are named **YYYYMMDD_da_expost_lmp.csv**. These files are available back to January 1, 2023.

- a. Within each file, use the row where the node is **DPC.Flambeau** and the value is **LMP**. Hourly prices are reported by hour ending; for example, the price for 1:00–2:00 p.m. is listed as HE 14.
4. For older data, open the MISO Market Report Archives page:
[https://www.misoenergy.org/markets-and-operations/real-time--market-data/market-report-archives/#nt=%2FMarketReportType%3AHistorical%20LMP%2FMarketReportName%3AArchived%20Day-Ahead%20Market%20ExPost%20LMPs%20%20\(zip\)&t=10&p=0&s=MarketReportPublished&sd=desc](https://www.misoenergy.org/markets-and-operations/real-time--market-data/market-report-archives/#nt=%2FMarketReportType%3AHistorical%20LMP%2FMarketReportName%3AArchived%20Day-Ahead%20Market%20ExPost%20LMPs%20%20(zip)&t=10&p=0&s=MarketReportPublished&sd=desc)
5. Confirm the highlighted folder on the left is **Archived Day-Ahead Market ExPost LMPs (zip)**.
6. Monthly archive files are named **YYYYMM_da_expost_lmp_csv.zip**. These archives contain data back to February 2015.
 - a. Each zip file contains daily CSV files in the same format described in 3a.

In the same folder as this problem description, you will also find **flambeau_gen.csv**. This file contains Flambeau's generation in MW, recorded on roughly a 4-day cadence, with some missing values.

Questions

Your presentation should answer the following questions:

1. How did you collect and prepare the LMP data? How did you merge it with the generation data?
2. What did you learn from an exploratory analysis of Flambeau's LMPs and generation?
3. What 2-week outage window do you recommend for 2027? Please include three possible outage date ranges, estimated lost revenue for each, and an uncertainty estimate for lost revenue.

Notes

We will try to load your emailed materials onto the conference room display before the interview. Please also bring a backup copy, such as your laptop, in case there is a technical issue.

You will be evaluated on:

1. The clarity and organization of your visuals.
2. Your ability to clearly explain your methodology and key findings.

3. Your ability to answer questions about your work.

Please keep the presentation to about 10 minutes. If you have any questions, feel free to contact David Elzinga at david.elzinga@dairylandpower.com.